

TECHNICAL SKILLS

Programming Languages: Java, Python, C/C++, Go, C#, SQL (PostgreSQL/MySQL), Bash/Shell Scripting
Libraries & Frameworks: PyTorch, scikit-learn, pandas, NumPy, OpenCV, Flask, FastAPI, Protocol Buffers
System & Infrastructure: Apache Kafka, Telegraf, InfluxDB, Grafana, Redfish, systemd
Cloud & DevOps: Git, GitLab CI/CD, Docker/Podman, Ansible, AWS
AI & Data: ChromaDB, RAG, ETL pipelines
Tools: GitHub Copilot, ChatGPT, opencode, Pytest, JUnit, LaTeX, Agile, Jira

EXPERIENCE

High Performance Computing (HPC) Engineering Intern

May 2026 – Present

Los Alamos National Laboratory

Los Alamos, NM

- Designed and deployed a hardware telemetry pipeline using Redfish API, Kafka, InfluxDB, and Telegraf, collecting 830K+ data points daily with zero data loss through containerized microservices and systemd automation.
- Reverse-engineered Intel Redfish API, implementing telemetry polling and event subscriptions across 6 endpoint families with a Python Flask receiver using dual-write pattern (Kafka + file storage).
- Provisioned a multi-node Rocky Linux cluster using OpenCHAMI, automating OS deployment via Kickstart and cloud-init, and implementing a GitLab CI/CD pipeline to automatically build and test boot images on commit.

Extended Reality (XR) Developer

July 2024 – May 2026

Mesh Labs (formerly Meteor Studio)

Mesa, AZ

- Containerized a GPU-enabled multi-dependency environment using Docker Compose, reducing setup time from 60 to 16 minutes and eliminating Linux/WSL2 compatibility issues.
- Developed a Python 3D camera calibration pipeline using RANSAC-based feature matching and ICP point cloud registration for accurate multi-view scene reconstruction.

Undergraduate Researcher

June 2023 – August 2023

Texas A&M-Corpus Christi

Corpus Christi, TX

- Developed multi-drone control system for PX4 drones in Gazebo simulation, debugging timing and communication across hardware/software boundaries while coordinating up to 10 autonomous drones.
- Designed windowing-based preprocessing pipelines for time-series sensor data, trained and validated KNN, SVM, Random Forest, and Naive Bayes ensemble models, improving gesture recognition accuracy by 10%.

PROJECTS

gRPC Microservices System

- Architected a distributed system with 5 gRPC-based microservices, implementing both interactive CLI and automated client interfaces with robust error handling and modular service design.
- Implemented 5 gRPC services (Fitness, RecipeBook, Sort, Joke, Echo) with Protocol Buffers, file-based persistence, a service registry for discovery, and thread-safe concurrent request handling.
- Deployed system on AWS EC2 instance with public accessibility, enabling multi-client access and validating concurrent service requests.

AI-Integrated Wireless Emergency Alerts

- Engineered a FastAPI microservice integrating retrieval-augmented generation (RAG) with ChromaDB to enable semantic search and context-aware LLM responses.
- Designed a containerized automation system with custom scheduling and health endpoints to orchestrate data ingestion, scraping, and LLM pipelines.
- Developed a fault-tolerant Python scraper for the PBS WARN API, utilizing exponential backoff to reliably extract and normalize live emergency data.
- Built a ChromaDB cleanup service that parses expiration timestamps and batch-deletes stale alerts on a 2-hour schedule, with a health endpoint verifying cleanup status.

Semantic Arithmetic Playground (detimzhao.dev)

- Built an interactive 3D vector-arithmetic explorer in Three.js with a formula parser, Levenshtein autocorrect, and 384-dimensional vector arithmetic.
- Engineered a Python pipeline converting 1,294 AI/ML concepts from HuggingFace, arXiv, Wikipedia, and PyTorch docs into 3D embeddings via PCA, where related ideas cluster.

EDUCATION

Bachelor of Science, Software Engineering

Arizona State University, Polytechnic

May 2026

Mesa, AZ